

MAT21-1

Linear Algebra II

Determinants, Eigenvalues, and Diagonalization

Finding the directions a matrix leaves unchanged.

Albert School — 34 hours

Preface

In Linear Algebra I you learned to **compute** with matrices: to multiply them, to solve $A\mathbf{x} = \mathbf{b}$ by Gaussian elimination, to read off a basis from the pivot columns, and to think of a matrix as a **linear map** f that sends a vector $\mathbf{x}$ to $A\mathbf{x}$. You also met the idea — only sketched there — that the **same** linear map looks like different matrices in different bases, and that a change of basis is recorded by an invertible matrix P .

This book is about a single question that ties all of that together:

Given a square matrix A , is there a basis in which it becomes as simple as possible — ideally just a list of scaling factors along the axes?

The simple matrices are the **diagonal** ones, because a diagonal matrix merely stretches each coordinate axis by a fixed amount. The special directions that get stretched without being rotated are the **eigenvectors**, the factors are the **eigenvalues**, and the process of rewriting A in a basis of eigenvectors is **diagonalization**. To decide when this is possible we first need a number that measures how a matrix distorts volume and detects invertibility: the **determinant**. Along the way we prove the surprising **Cayley–Hamilton theorem** and put everything to work on walks in graphs and on Markov chains.

Everything rests on the prerequisites from Linear Algebra I (MAT11-3): matrix arithmetic, Gaussian elimination, vector spaces and subspaces, bases and dimension, the rank theorem, and coordinate vectors under a change of basis. We build on those rather than repeating them; if a term like **kernel**, **rank**, or **basis** feels unfamiliar, revisit that course first.

Throughout, $\mathbb{R}$ is the real numbers, $\mathbb{C}$ the complex numbers, $M_n(\mathbb{R})$ the $n \times n$ real matrices, and I_n (or just I) the identity matrix. We work with real matrices but occasionally allow complex eigenvalues, because — as you will see — some perfectly ordinary real matrices can only be diagonalized over $\mathbb{C}$.

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1 Determinants

The determinant attaches to every square matrix A a single number $\det(A)$. That number answers a yes/no question we care about constantly — **is A invertible?** — and, more than that, measures **how much A stretches or collapses volume**. We start with the smallest cases, where you can see the geometry directly, then build the general machinery.

1.1 A 2×2 example you can see

Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

As a linear map it sends the standard basis vectors to its columns: $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ and $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 3 \end{pmatrix}$. The unit square spanned by e_1 and e_2 — which has area 1 — is sent to the parallelogram spanned by those two image vectors.

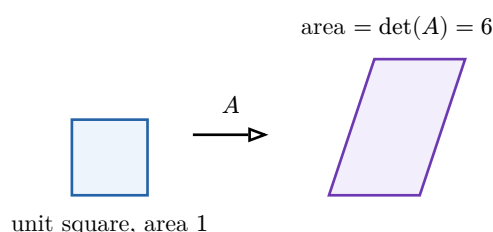


Figure 1: The map A sends the unit square to a parallelogram whose area is $\det(A)$.

The area of that parallelogram turns out to be exactly $2 \cdot 3 - 1 \cdot 0 = 6$. That number is the determinant. The general rule for 2×2 matrices is

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Worked example. $\det \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} = 2 \cdot 3 - 1 \cdot 0 = 6$, matching the picture. For $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ we get $1 \cdot 4 - 2 \cdot 2 = 0$: the two columns are proportional, the “parallelogram” is flat, and indeed this matrix is not invertible.

The sign matters too. If the determinant is negative, the map has **flipped orientation** (it turns a counter-clockwise corner into a clockwise one), and $|\det(A)|$ is the area. This is why we speak of the determinant as a **signed** area — in higher dimensions, a **signed volume**.

Definition 1 (determinant, the property that pins it down) The determinant is the unique function $\det : M_n(\mathbb{R}) \rightarrow \mathbb{R}$ that is

1. **linear in each column separately** (with the other columns held fixed),
2. **zero whenever two columns are equal**, and
3. **normalized** so that $\det(I_n) = 1$.

Geometrically, $\det(A)$ is the signed n -dimensional volume of the parallelepiped spanned by the columns of A .

We will not prove uniqueness, but every computational rule below follows from these three properties.

1.2 The 3×3 case: Sarrus' rule

For 3×3 matrices there is a convenient mnemonic. Write the matrix and **repeat the first two columns** to its right:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}.$$

Then add the three “down-right” diagonal products and subtract the three “down-left” ones:

$$\det = aei + bfg + cdh - ceg - afh - bdi.$$

Worked example — Sarrus.

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix} = (1 \cdot 1 \cdot 0 + 2 \cdot 4 \cdot 5 + 3 \cdot 0 \cdot 6) - (3 \cdot 1 \cdot 5 + 1 \cdot 4 \cdot 6 + 2 \cdot 0 \cdot 0).$$

The first group is $0 + 40 + 0 = 40$; the second is $15 + 24 + 0 = 39$. So $\det = 40 - 39 = 1$.

Common pitfall Sarrus' rule works **only** for 3×3 matrices. There is no analogous “extend the columns” trick for 4×4 or larger — a mistake students make constantly. For bigger matrices use cofactor expansion or row reduction below.

1.3 Cofactor expansion (Laplace expansion)

The general-purpose definition expresses an $n \times n$ determinant in terms of smaller ones. Let A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained by **deleting row i and column j** of A . The number

$$C_{ij} = (-1)^{i+j} \det(A_{ij})$$

is the **cofactor** of the entry a_{ij} ; the sign $(-1)^{i+j}$ follows the checkerboard pattern

$$\begin{pmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Theorem 2 (cofactor / Laplace expansion) For any row i ,

$$\det(A) = \sum_{j=1}^n a_{ij}C_{ij},$$

and likewise the expansion along any column j gives $\det(A) = \sum_{i=1}^n a_{ij}C_{ij}$. Every row and every column yields the same value.

Because you may expand along **any** row or column, choose the one with the most zeros to save work.

Worked example — expanding along a sparse column. Take

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 6 & 0 \end{pmatrix}.$$

The first column has two zeros, so expand along it. Only $a_{11} = 1$ survives:

$$\det(A) = 1 \cdot (-1)^{1+1} \det \begin{pmatrix} 1 & 4 \\ 6 & 0 \end{pmatrix} = 1 \cdot (1 \cdot 0 - 4 \cdot 6) = -24.$$

One step, instead of six products with Sarrus.

Common pitfall Do not forget the sign factor $(-1)^{i+j}$. Expanding along the **second** row or column starts with a **minus**. Writing the checkerboard in the margin before you begin prevents the single most common determinant error.

1.4 Determinants by row reduction

Cofactor expansion costs roughly $n!$ multiplications — hopeless for large n . The efficient method is to run Gaussian elimination while **tracking what each operation does to the determinant**. The three elementary row operations behave as follows.

Proposition 3 (effect of row operations)

1. **Swapping two rows** multiplies the determinant by -1 .
2. **Multiplying a row by a scalar k** multiplies the determinant by k .
3. **Adding a multiple of one row to another** leaves the determinant unchanged.

The third operation is the workhorse: it changes nothing, so we may freely reduce A to an upper-triangular matrix U . And triangular determinants are trivial:

Proposition 4 (triangular matrices) If A is upper- or lower-triangular, $\det(A)$ is the product of its diagonal entries $a_{11}a_{22}\dots a_{nn}$.

Worked example — row reduction. Reduce

$$A = \begin{pmatrix} 2 & 4 & 2 \\ 1 & 3 & 2 \\ 1 & 1 & 3 \end{pmatrix}.$$

Subtract $\frac{1}{2}R_1$ from R_2 and from R_3 (operation 3, no change):

$$\rightarrow \begin{pmatrix} 2 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Add R_2 to R_3 (no change):

$$\rightarrow \begin{pmatrix} 2 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

This is triangular, so $\det(A) = 2 \cdot 1 \cdot 3 = 6$. We used only operation 3, so no correction factor is needed.

Common pitfall If you **do** swap rows or scale a row, you must compensate. A swap flips the sign; scaling a row by k means the triangular product must be divided by k to recover $\det(A)$. Forgetting these corrections is the classic row-reduction determinant bug.

1.5 Properties that save computation

These follow from the defining properties and are worth memorizing.

Proposition 5 (key properties) For square matrices A, B of the same size:

1. **Multiplicativity:** $\det(AB) = \det(A)\det(B)$.
2. **Transpose:** $\det(A^T) = \det(A)$ — so every row statement has a column twin.
3. **Invertibility criterion:** A is invertible $\Leftrightarrow \det(A) \neq 0$, and then $\det(A^{-1}) = 1/\det(A)$.
4. **Triangular / diagonal:** the determinant is the product of the diagonal.
5. **Scaling:** $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix (each of the n columns contributes a factor k).

Worked example — multiplicativity in action. If $\det(A) = 3$ and $\det(B) = -2$ then $\det(AB) = -6$, $\det(A^2) = 9$, and if A is 4×4 then $\det(2A) = 2^4 \cdot 3 = 48$ — **not** $2 \cdot 3$.

The invertibility criterion is the bridge to the rest of the book: the whole theory of eigenvalues hinges on choosing a number λ that makes a certain matrix **non**-invertible, i.e. makes its determinant zero.

Exercises

1. Compute $\det \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$ and $\det \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 0 & 4 & 1 \end{pmatrix}$ (the second by Sarrus and again by cofactor expansion; check they agree).
2. Use row reduction to compute $\det \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$, stating which operations you use and any sign corrections.

3. Given $\det(A) = 5$ for a 3×3 matrix, find $\det(2A)$, $\det(A^T)$, $\det(A^{-1})$, and $\det(A^3)$.
4. Show that if A has a row of zeros then $\det(A) = 0$, two ways: by cofactor expansion along that row, and by the invertibility criterion.
5. A matrix Q satisfies $Q^T Q = I$. Show that $\det(Q) = \pm 1$.

2 Eigenvalues and Eigenvectors

We now reach the heart of the course. An eigenvector of A is a nonzero direction that the map $\mathbf{x} \rightarrow A\mathbf{x}$ does not turn — it only scales. The scaling factor is the eigenvalue. These special directions expose the **intrinsic** behaviour of a matrix, independent of the coordinate system you happened to write it in.

2.1 A first example

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Most vectors get rotated when you apply A . But try $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$:

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The output points the **same way** as the input, just three times longer. And try $\mathbf{w} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$:

$$A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

This one is left completely fixed. These two directions are **eigenvectors**, with **eigenvalues** 3 and 1.

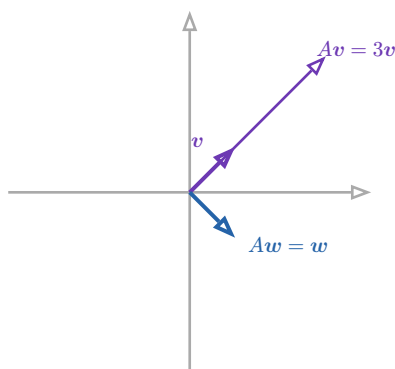


Figure 2: Eigenvectors of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: the direction $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is stretched by 3, the direction $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is fixed.

2.2 Definitions

Definition 6 (eigenvalue, eigenvector, eigenspace) Let $A \in M_n(\mathbb{R})$. A scalar λ is an **eigenvalue** of A if there is a **nonzero** vector $\mathbf{v}$ with

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Such a $\mathbf{v}$ is an **eigenvector** associated with λ . The set

$$E_\lambda = \{v : Av = \lambda v\} = \ker(A - \lambda I)$$

of all such vectors (including $\mathbf{0}$) is the **eigenspace** of λ ; it is a subspace of $\mathbb{R}^n$.

Common pitfall An **eigenvector** must be nonzero — otherwise $A\mathbf{0} = \lambda\mathbf{0}$ would hold for every λ and the notion would be useless. An **eigenvalue**, by contrast, *may* be 0 (that happens exactly when A is not invertible). And the **eigenspace** E_λ , being a kernel, *does* contain $\mathbf{0}$.

2.3 The characteristic polynomial

How do we **find** the eigenvalues, rather than guessing? Rewrite the defining equation:

$$Av = \lambda v \Leftrightarrow (A - \lambda I)v = \mathbf{0}.$$

We want a **nonzero** v in the kernel of $A - \lambda I$. By the invertibility criterion of Chapter 1, $A - \lambda I$ has a nonzero kernel exactly when it is **not** invertible, i.e. when its determinant vanishes.

Theorem 7 (characteristic equation) λ is an eigenvalue of $A \Leftrightarrow \det(A - \lambda I) = 0$. The function

$$\chi_A(\lambda) = \det(A - \lambda I)$$

is a polynomial of degree n in λ , the **characteristic polynomial** of A . Its roots are exactly the eigenvalues.

Worked example — from matrix to eigenvalues. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$,

$$\chi_A(\lambda) = \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The roots $\lambda = 1$ and $\lambda = 3$ are the eigenvalues we found by hand. To get the eigenspace for $\lambda = 3$, solve $(A - 3I)v = \mathbf{0}$:

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x = y,$$

so $E_3 = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$. Similarly $E_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$.

Common pitfall Compute $\det(A - \lambda I)$, **not** $\det(\lambda I - A)$, or your polynomial's sign may flip on odd-sized matrices. The **roots** are the same either way, so many books use $\det(\lambda I - A)$ to keep the leading term $+\lambda^n$; just be consistent. Also: always subtract λ from the **diagonal only** — a surprisingly common slip is to subtract it from every entry.

2.4 Trace and determinant from the eigenvalues

Two coefficients of the characteristic polynomial have a beautiful meaning. The **trace** of A , written $\text{tr}(A)$, is the sum of its diagonal entries.

Proposition 8 (trace and determinant) If $A \in M_n$ has eigenvalues $\lambda_1, \dots, \lambda_n$ (listed with algebraic multiplicity, possibly complex), then

$$\operatorname{tr}(A) = \lambda_1 + \dots + \lambda_n \quad \text{and} \quad \det(A) = \lambda_1 \lambda_2 \dots \lambda_n.$$

Worked example. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $\operatorname{tr}(A) = 4 = 1 + 3 \checkmark$ and $\det(A) = 3 = 1 \cdot 3 \checkmark$. These two equations are a fast sanity check: after computing eigenvalues, verify their sum is the trace and their product is the determinant before going further.

2.5 When eigenvalues go complex

Not every real matrix has real eigenvalues. The rotation by 90° ,

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

turns **every** direction, so geometrically no real eigenvector can exist. Indeed

$$\chi_R(\lambda) = \lambda^2 + 1,$$

whose roots are $\lambda = \pm i$. Over $\mathbb{C}$ the eigenvalues exist; over $\mathbb{R}$ they do not. This is why complex numbers enter linear algebra, and why some matrices are diagonalizable only over $\mathbb{C}$ — a theme of the next chapter.

Exercises

6. Find the eigenvalues and a basis of each eigenspace for $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.
7. Compute the characteristic polynomial of $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 2 & -4 & 1 \end{pmatrix}$ and find all eigenvalues. (Hint: expand along a sparse row or column.)
8. Verify the trace/determinant relations for the matrix in the previous exercise.
9. Show that A and A^T have the same characteristic polynomial, hence the same eigenvalues. (Use $\det(M) = \det(M^T)$.)
10. Find the eigenvalues of $\begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$ and explain why the eigenvectors are not real.

3 Diagonalization

We can now answer the book's central question. A matrix is **diagonalizable** when it has **enough** independent eigenvectors to form a basis. In that basis the map is just a diagonal scaling — and that makes powers, recurrences, and long-run behaviour easy.

3.1 Two kinds of multiplicity

A repeated eigenvalue can behave in two different ways, and telling them apart is the whole game.

Definition 9 (algebraic and geometric multiplicity) Let λ be an eigenvalue of A .

- Its **algebraic multiplicity** $m_a(\lambda)$ is its multiplicity as a root of the characteristic polynomial χ_A .
- Its **geometric multiplicity** $m_g(\lambda) = \dim E_\lambda$ is the number of independent eigenvectors it carries.

These always satisfy $1 \leq m_g(\lambda) \leq m_a(\lambda)$.

Worked example — when geometric < algebraic. The shear

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

has $\chi_A(\lambda) = (2 - \lambda)^2$, so $\lambda = 2$ with $m_a = 2$. But

$$A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has rank 1, so $\dim E_2 = 2 - 1 = 1$: only **one** independent eigenvector, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Here $m_g = 1 < 2 = m_a$. This shortfall is exactly what will block diagonalization.

Common pitfall A repeated eigenvalue does **not** automatically mean trouble. The identity matrix I_2 has $\lambda = 1$ with $m_a = 2$ **and** $m_g = 2$ — every vector is an eigenvector. What blocks diagonalization is specifically $m_g < m_a$, the **deficient** case, not repetition itself.

3.2 The diagonalization criterion

Theorem 10 (diagonalizability criterion) A matrix $A \in M_n$ is **diagonalizable** (over the field in use) if and only if both:

1. its characteristic polynomial **splits** into linear factors over that field (all n roots lie in it, counted with multiplicity); and
2. for **every** eigenvalue, $m_g(\lambda) = m_a(\lambda)$.

Equivalently: A is diagonalizable $\Leftrightarrow$ it has n linearly independent eigenvectors, which together form an eigenbasis of the space.

A useful shortcut falls out immediately.

Corollary 11 (distinct eigenvalues) If an $n \times n$ matrix has n **distinct** eigenvalues, it is automatically diagonalizable, because each contributes one independent eigenvector and eigenvectors for different eigenvalues are independent.

3.3 What “ $A = PDP^{-1}$ ” means

Suppose A has an eigenbasis $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$. Place the eigenvectors as the **columns** of a matrix P and the eigenvalues on the diagonal of D :

$$P = (\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n), \quad D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.$$

Then $AP = PD$ (column j of each side is $A\mathbf{v}_j = \lambda_j\mathbf{v}_j$), and since the independent columns make P invertible,

$$A = PDP^{-1}.$$

This is precisely the **change of basis** from Linear Algebra I: P converts eigen-coordinates to standard coordinates, D does the scaling, P^{-1} converts back. A and D are **similar** matrices — the same map in two bases.

Worked example — a full diagonalization. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ we found $\lambda_1 = 1$ with $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\lambda_2 = 3$ with $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. So

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \quad P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

and one checks $A = PDP^{-1}$.

Remark The columns of P and the diagonal of D must be in the **same order**: the j -th column of P must be an eigenvector for the j -th diagonal entry of D . Swapping one without the other is a frequent and silent error.

3.4 Why diagonalize: powers and recurrences

The payoff is that **powers become trivial**. Since

$$A^k = (PDP^{-1})(PDP^{-1})\dots(PDP^{-1}) = PD^kP^{-1},$$

and D^k just raises each diagonal entry to the k -th power,

$$D^k = \begin{pmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{pmatrix}.$$

Computing A^{100} directly would take dozens of matrix multiplications; through diagonalization it is three multiplications and n scalar powers.

Worked example — Fibonacci via diagonalization. The Fibonacci numbers $F_0 = 0, F_1 = 1, F_{k+1} = F_k + F_{k-1}$ satisfy

$$\begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_k \\ F_{k-1} \end{pmatrix}, \quad \text{so} \quad \begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = A^k \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Here $\chi_A(\lambda) = \lambda^2 - \lambda - 1$ with roots $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Diagonalizing and reading off the top entry of $A^k \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ gives the closed form

$$F_k = \frac{\varphi^k - \psi^k}{\sqrt{5}}.$$

A recurrence, solved by an eigenvalue computation: this is the template for every linear recurrence.

3.5 A non-diagonalizable matrix

Return to the shear $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Its only eigenvalue $\lambda = 2$ has $m_g = 1 < 2 = m_a$, so there is no eigenbasis and A is **not** diagonalizable. We cannot write it as PDP^{-1} ; the closest simple form is the **triangular** shape it already has. The next chapter gives a tool — Cayley–Hamilton — that handles powers of such matrices anyway.

Exercises

11. Diagonalize $\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$: find P and D , and verify $A = PDP^{-1}$.
12. For each, decide diagonalizability and justify with multiplicities: $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$, $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}$.
13. Using diagonalization, compute A^k in closed form for $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
14. Solve the recurrence $a_{k+1} = 3a_k - 2a_{k-1}$ with $a_0 = 1, a_1 = 3$ by writing it in matrix form and diagonalizing.
15. Show that if A is diagonalizable then so is A^2 , and its eigenvalues are the squares of those of A .

4 The Cayley–Hamilton Theorem

Here is one of the most striking facts in the subject: **every square matrix satisfies its own characteristic equation**. This lets us compute inverses and high powers — even of matrices that cannot be diagonalized.

4.1 Statement and a worked check

Theorem 12 (Cayley–Hamilton) Let $\chi_A(\lambda) = \det(A - \lambda I)$ be the characteristic polynomial of $A \in M_n$. Then substituting the matrix A for λ gives the zero matrix:

$$\chi_A(A) = \mathbf{0}.$$

(The constant term c_0 of χ_A becomes $c_0 I$ when substituted.)

Common pitfall Cayley–Hamilton is **not** the triviality “ $\det(A - A \cdot I) = \det(0) = 0$ ”. That manipulation is meaningless: χ_A is a **polynomial in a scalar**, and we are substituting a **matrix** into it, producing a matrix equation. You must build $\chi_A(\lambda) = \lambda^n + \dots + c_0$ first, **then** replace each λ^k by A^k and c_0 by $c_0 I$.

Worked example — verifying Cayley–Hamilton. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ we had $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$. Cayley–Hamilton claims $A^2 - 4A + 3I = \mathbf{0}$. Check:

$$A^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}, \quad A^2 - 4A + 3I = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} - \begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \checkmark$$

4.2 Application 1: the inverse as a polynomial in A

Rearranging the characteristic equation isolates the identity, and dividing by the (nonzero) constant term expresses A^{-1} using only matrix powers of A — no Gaussian elimination needed.

Worked example — inverse via Cayley–Hamilton. With $A^2 - 4A + 3I = \mathbf{0}$, move terms and factor out A :

$$A^2 - 4A = -3I \quad \Rightarrow \quad A(A - 4I) = -3I \quad \Rightarrow \quad A \cdot \underbrace{\left(-\frac{1}{3}\right)(A - 4I)}_{=A^{-1}} = I.$$

So

$$A^{-1} = -\frac{1}{3}(A - 4I) = -\frac{1}{3}\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

In general, if $\chi_A(\lambda) = \lambda^n + \dots + c_1\lambda + c_0$ with $c_0 = \det(A) \neq 0$, then

$$A^{-1} = -\frac{1}{c_0}(A^{n-1} + \dots + c_1I).$$

Remark This needs $c_0 = (-1)^n \det(A) \neq 0$, i.e. A invertible — consistent with 0 not being an eigenvalue. If $\det(A) = 0$ the constant term vanishes and the trick instead expresses a **power** of A in terms of lower powers.

4.3 Application 2: powers of a non-diagonalizable matrix

Cayley–Hamilton says A^n equals a combination of $I, A, \dots, A^{n-1}$. Hence **every** power A^k — however large k — reduces to such a combination. This works even when A is **not** diagonalizable, the case the previous chapter left open.

Worked example — reducing high powers. For the shear $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, $\chi_A(\lambda) = (\lambda - 2)^2 = \lambda^2 - 4\lambda + 4$, so by Cayley–Hamilton

$$A^2 = 4A - 4I.$$

Every higher power now collapses to the form $\alpha A + \beta I$. For instance

$$A^3 = A \cdot A^2 = A(4A - 4I) = 4A^2 - 4A = 4(4A - 4I) - 4A = 12A - 16I,$$

giving $A^3 = \begin{pmatrix} 8 & 12 \\ 0 & 8 \end{pmatrix}$ — confirmed by direct multiplication, but obtained without it, and applicable to a matrix that has no diagonalization.

4.4 Computing the inverse via Gaussian elimination

For completeness we record the standard **direct** route to A^{-1} , the counterpart to the Cayley–Hamilton shortcut. Form the augmented matrix $[A \mid I]$ and row-reduce until the left block becomes I ; the right block is then A^{-1} :

$$[A \mid I] \xrightarrow{\text{row reduce}} [I \mid A^{-1}].$$

Worked example — inverse by elimination.

$$[A \mid I] = \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right).$$

Swap rows, eliminate, and scale to reach $\left(\begin{array}{cc|cc} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} \end{array} \right)$, so $A^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ — the same answer Cayley–Hamilton gave, as it must.

Exercises

16. Verify Cayley–Hamilton for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

17. Use Cayley–Hamilton to compute A^{-1} for $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$, then check it by Gaussian elimination on $[A \mid I]$.

18. For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, use Cayley–Hamilton to express A^k in the form $\alpha A + \beta I$ and find A^5 .
19. Let A be 3×3 with $\chi_A(\lambda) = -\lambda^3 + 2\lambda^2 + \lambda - 2$. Write A^{-1} as a polynomial in A .
20. Explain why Cayley–Hamilton lets you express **any** power A^k (for $k \geq n$) as a polynomial in A of degree at most $n - 1$.

5 Applications: Graphs and Markov Chains

The abstract machinery now pays dividends. Two settings — counting walks in a network and predicting the long-run behaviour of a random process — both reduce to matrix powers, and both are illuminated by eigenvalues and diagonalization.

5.1 Walks in a graph via A^k

Model a network (web pages, friendships, road links) by its **adjacency matrix**: number the vertices $1, \dots, n$ and set $a_{ij} = 1$ if there is an edge from i to j , else 0.

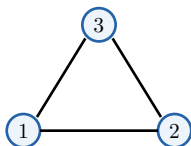


Figure 3: A triangle graph on three vertices; every pair is joined by an edge.

The fundamental fact is that **matrix multiplication counts walks**.

Theorem 13 (walk counting) If A is the adjacency matrix of a graph, then the (i, j) entry of A^k is the number of **walks of length k** from vertex i to vertex j .

The reason is exactly the definition of matrix multiplication: the (i, j) entry of $A^2 = A \cdot A$ is $\sum_m a_{im}a_{mj}$, which counts the intermediate vertices m giving a length-2 walk $i \rightarrow m \rightarrow j$. Induction extends this to every k .

Worked example — counting paths. For the triangle above,

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

The diagonal 2's say there are two walks of length 2 from a vertex back to itself (out to either neighbour and back); the off-diagonal 1's count the single length-2 route between any two vertices (via the third). Multiply again for A^3 to count length-3 walks.

5.2 The dominant eigenvalue governs growth

How fast does the number of walks grow as $k \rightarrow \infty$? Diagonalize $A = PDP^{-1}$; then $A^k = PD^kP^{-1}$, and as k grows the **largest** eigenvalue $\lambda_{\max}$ dwarfs the others. The entries of A^k grow like $\lambda_{\max}^k$, and the **direction** of growth is the eigenvector of $\lambda_{\max}$.

The general principle. For any linear iteration $\mathbf{x}_{k+1} = A\mathbf{x}_k$, expand the start vector in an eigenbasis, $\mathbf{x}_0 = c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n$. Then

$$\mathbf{x}_k = A^k\mathbf{x}_0 = c_1\lambda_1^k\mathbf{v}_1 + \dots + c_n\lambda_n^k\mathbf{v}_n.$$

The term with the largest $|\lambda|$ eventually dominates, so $\mathbf{x}_k$ aligns with that **dominant eigenvector** and scales by $\lambda_{\max}$ per step. Eigenvalues are the asymptotics of iteration.

5.3 Markov chains and stochastic matrices

A **Markov chain** is a random process that hops between states, where the next state depends only on the current one. Collect the transition probabilities into a matrix. We use the **column-stochastic** convention: p_{ij} is the probability of moving **from state j to state i** , so **each column sums to 1**. A probability distribution is a column vector $\mathbf{x}$ with nonnegative entries summing to 1, and one step of the chain is $\mathbf{x}_{k+1} = P\mathbf{x}_k$.

Worked example — a weather chain. States: sunny (S) and rainy (R). If today is sunny, tomorrow is sunny with probability 0.8; if rainy, sunny with probability 0.4:

$$P = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}, \quad \text{columns sum to } 1.$$

Starting from a sunny day $\mathbf{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, the distribution after k days is $P^k \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Common pitfall Watch the convention. With **column-stochastic** P you multiply on the **left**, $\mathbf{x}_{k+1} = P\mathbf{x}_k$, and columns sum to 1. Many texts use **row-stochastic** matrices with row vectors $\mathbf{x}_{k+1} = \mathbf{x}_k P$ and rows summing to 1. Both are correct; mixing them is the usual source of wrong answers. Pick one and check that your “sum to 1” axis matches your multiplication side.

5.4 The stationary distribution is an eigenvector for $\lambda = 1$

A distribution $\boldsymbol{\pi}$ is **stationary** if one step leaves it unchanged: $P\boldsymbol{\pi} = \boldsymbol{\pi}$. That is **exactly** the eigenvector equation for eigenvalue $\lambda = 1$.

Proposition 14 (eigenvalue 1) Every column-stochastic matrix P has $\lambda = 1$ as an eigenvalue, and a stationary distribution is an eigenvector for it (rescaled so its entries are nonnegative and sum to 1). All eigenvalues satisfy $|\lambda| \leq 1$.

The reason 1 is always an eigenvalue: since every column of P sums to 1, every column of $P - I$ sums to 0, so the rows of $P - I$ are linearly dependent, making $\det(P - I) = 0$.

Worked example — stationary weather. Solve $(P - I)\boldsymbol{\pi} = \mathbf{0}$ for the weather chain:

$$\begin{pmatrix} -0.2 & 0.4 \\ 0.2 & -0.4 \end{pmatrix} \begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \pi_1 = 2\pi_2.$$

Normalizing $\pi_1 + \pi_2 = 1$ gives $\boldsymbol{\pi} = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$: in the long run it is sunny two-thirds of the time, **regardless of today’s weather**.

5.5 Convergence to stationarity via diagonalization

Why does the chain **forget** its starting point? Diagonalize $P = QDQ^{-1}$. The eigenvalue 1 contributes a term that never decays — the stationary distribution — while every other eigenvalue has $|\lambda| < 1$, so its contribution $\lambda^k \rightarrow 0$. Writing the start vector in the eigenbasis,

$$\mathbf{x}_k = P^k \mathbf{x}_0 = \underbrace{\boldsymbol{\pi}}_{\lambda=1 \text{ term}} + \sum_{|\lambda_j| < 1} c_j \lambda_j^k \mathbf{v}_j \rightarrow \boldsymbol{\pi}.$$

Every transient term dies; what remains is the stationary distribution. The **rate** of convergence is set by the second-largest $|\lambda|$ — the closer it is to 1, the slower the chain mixes.

Worked example — watching it converge. The weather matrix has eigenvalues 1 and 0.4 (their sum is the trace 1.4). Starting sunny:

$$\mathbf{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{x}_1 = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 0.72 \\ 0.28 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 0.688 \\ 0.312 \end{pmatrix} \rightarrow \begin{pmatrix} 0.667 \\ 0.333 \end{pmatrix}.$$

The gap to $\boldsymbol{\pi}$ shrinks by the factor 0.4 each day — the second eigenvalue made visible.

Exercises

21. For the path graph $1 - 2 - 3$ (edges 1-2 and 2-3 only), write the adjacency matrix and compute A^2 . How many length-2 walks join vertex 1 to vertex 3?
22. A column-stochastic matrix $P = \begin{pmatrix} 0.5 & 0.3 \\ 0.5 & 0.7 \end{pmatrix}$ models customer switching between two brands. Find its stationary distribution.
23. Verify that $\lambda = 1$ is an eigenvalue of any 3×3 column-stochastic matrix by showing the columns of $P - I$ sum to zero.
24. For the weather chain, use diagonalization to find a closed form for $\mathbf{x}_k$ and confirm it converges to $\begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$.
25. Explain, using the dominant-eigenvalue principle, why the number of walks of length k in a connected graph grows roughly like $\lambda_{\max}^k$ where $\lambda_{\max}$ is the largest eigenvalue of the adjacency matrix.

Where we have arrived. Starting from the determinant — a single number measuring volume change and detecting invertibility — we built the characteristic polynomial, whose roots are the eigenvalues, the directions a matrix only scales. Matching geometric to algebraic multiplicity decides diagonalizability, $A = PDP^{-1}$, which turns powers and recurrences into scalar arithmetic. Cayley–Hamilton extends the reach to non-diagonalizable matrices and to inverses. Finally, walks in graphs and Markov chains revealed the same lesson: **the long-run behaviour of any linear iteration is governed by its dominant eigenvalue and the eigenvector pointing along it.** The change of basis you met in Linear Algebra I was, all along, the search for the coordinates in which a transformation is simplest.